Improving Emotional Text-to-Speech in Low-Resource Languages via Reinforcement Learning

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Abstract

We address emotional capability loss when an expressive text-to-speech (TTS) model trained on a high-resource language is adapted to a low-resource language. Using IndexTTS2 as the expressive base model, we propose an RL-enhanced adaptation framework for preserving emotional controllability, prosodic richness, and speech naturalness during transfer from Chinese to Taiwanese. The goal is to retain the pretrained emotional prior and realign it to limited target-language data.

Motivation and Objectives

Expressive low-resource TTS is limited by a tension between pronunciation adaptation and emotion preservation. Naive fine-tuning can improve intelligibility while flattening rhythm, pauses, emphasis, and intensity.

- **Emotional forgetting:** target-language fine-tuning can erase expressive behavior learned from high-resource speech.
- **Prosodic control loss:** adapted speech may stay intelligible but lose natural emphasis and intensity control.
- **Cross-lingual alignment:** emotional cues do not transfer perfectly across languages with different phonology and rhythm.
- **Objective:** recover controllable emotional expression in Taiwanese TTS through RL-based realignment.

Proposed Framework

The method first adapts the expressive base model to Taiwanese pronunciation, then applies RL to recover controllable emotion and prosody.

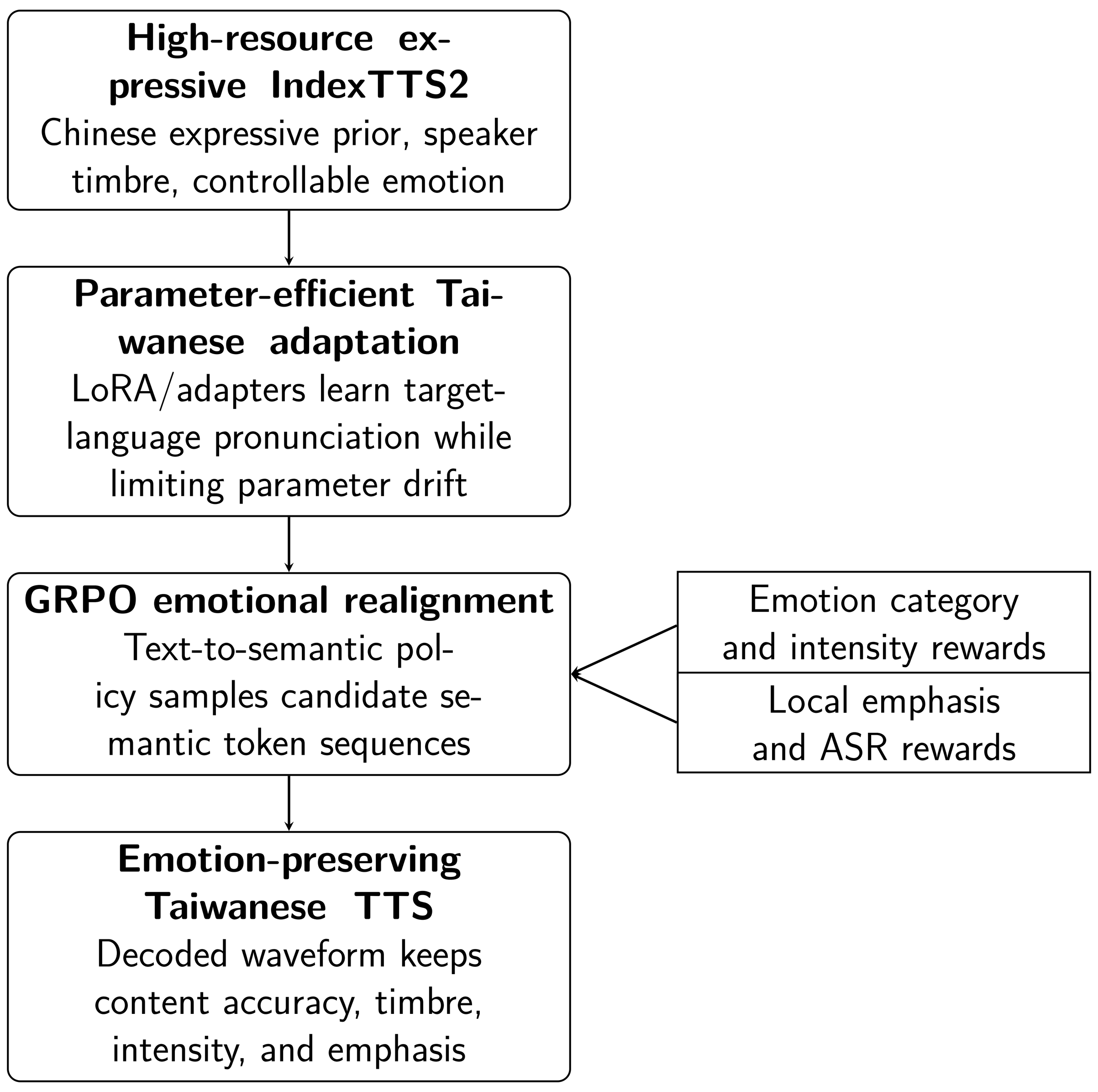

Figure 1. Two-stage adaptation and RL realignment pipeline.

- **Stage 1:** LoRA or adapter-based adaptation limits parameter drift and reduces catastrophic forgetting.
- **Stage 2:** GRPO optimizes the autoregressive text-to-semantic policy with waveform-level rewards.
- **Anchor:** a KL penalty preserves pronunciation, timbre, and the adapted reference behavior.

RL Formulation

Semantic speech generation is formulated as an MDP over the IndexTTS2 text-to-semantic module.

- **State:** input text, speaker timbre, duration constraint, target emotion, and prior semantic tokens.
- **Action:** the next discrete semantic speech token from policy π_θ.
- **Reward:** a sequence-level score computed from the decoded waveform.

$$R = \lambda_1 R_{\text{emo}} + \lambda_2 R_{\text{intensity}} + \lambda_3 R_{\text{emphasis}} + \lambda_4 R_{\text{ASR}}$$

$$\max_{\theta} \; \mathbb{E}[R(s,a)] - \beta \, \mathrm{KL}(\pi_\theta(\cdot|s) \,\|\, p_{\text{adapt}}(\cdot|s))$$

Datasets and Baselines

The experiment is designed to separate basic language adaptation from emotional recovery, so each model is evaluated before adaptation, after low-resource adaptation, and after RL realignment.

- **Emotion supervision:** ESD and Expresso provide emotion labels, intensity cues, and emphasis annotations.
- **Taiwanese adaptation:** TAT_MOE supports target-language adaptation and evaluation with low-resource subsets of roughly 1–5 hours.
- **Baselines:** high-resource IndexTTS2, naive fine-tuning, parameter-efficient adaptation, our RL-enhanced model, and CosyVoice3.
- **Core comparison:** Emotion Retention Gap and recovery ratio measure how much expressiveness is lost and then restored.

Architecture and Reward Design

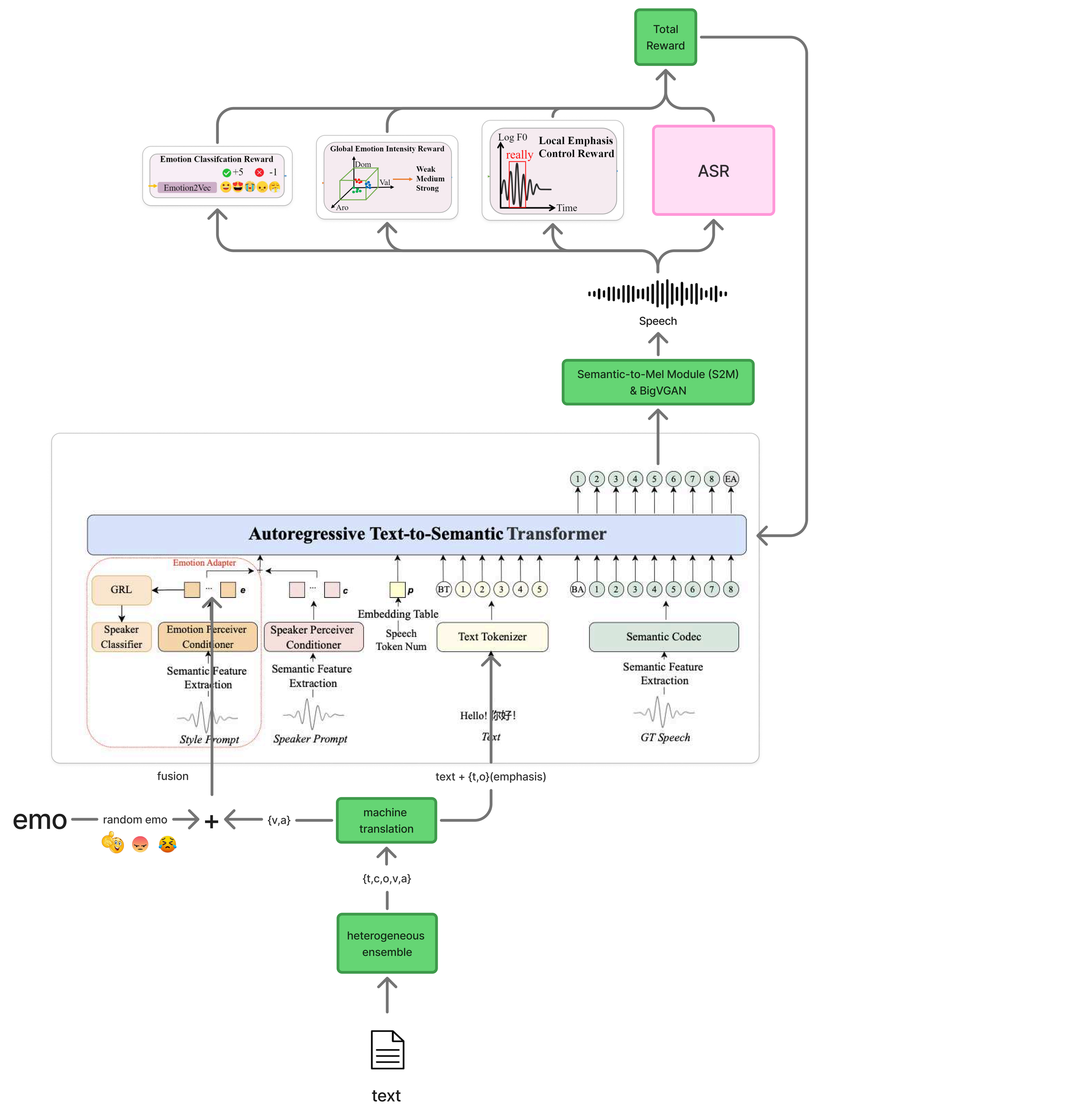

Figure 2. RL-enhanced IndexTTS2 architecture with waveform-level reward models.

Reward models evaluate decoded speech rather than only token likelihood.

- **Emotion category:** emotion2vec checks target emotion match.
- **Global intensity:** VAD distance from a neutral centroid encourages stronger affect.
- **Local emphasis:** pitch and energy patterns measure stressed words and expressive prosody.
- **Content accuracy:** ASR transcription compares generated speech with the input text.

Evaluation Protocol

We evaluate intelligibility, speaker fidelity, emotional expressiveness, and prosodic control across three stages: source model, adapted model, and RL-optimized model.

- **Objective metrics:** WER from ASR, speaker similarity, emotion similarity or accuracy, Emotion Retention Gap, and recovery ratio.
- **Subjective MOS:** QMOS, SMOS, PMOS, and EMOS on a 1–5 scale for quality, speaker identity, prosody, and emotion match.
- **Fine-grained tests:** Emotion Intensity Test compares weak, medium, and strong generations; Emphasis Accuracy Test checks perceived stress locations.
- **Trade-off check:** gains in emotion should not degrade WER, speaker similarity, or naturalness.

Preliminary Taiwanese TTS Baseline

A pilot MOS evaluation with 12 SARC laboratory members used 20 sentences for intelligibility and naturalness plus 10 sentences for voice similarity, yielding 600 subjective rating samples.

Table 1. Preliminary subjective evaluation (MOS) on Taiwanese TTS.

Model	Similarity MOS ↑	Intelligibility MOS ↑	Naturalness MOS ↑	Mean ↑
CosyVoice2	4.16	4.41	4.37	4.31

CosyVoice2 reaches a mean MOS of 4.31, motivating a shift from basic quality toward emotional preservation and recovery.

Expected Contributions

- A low-resource emotional TTS adaptation framework separating pronunciation transfer from expressive realignment.
- A multi-objective reward design for emotion, intensity, emphasis, and semantic intelligibility.
- An evaluation protocol for emotional forgetting, recovery ratio, speaker similarity, and WER trade-offs.

Team and References

Team: Hao-Chun Hsieh – formulation and methodology; Pao-Hua Hsu – documentation, visualization, poster, and presentation; Nien-Tse Wu – data, evaluation, implementation, and tuning.

References: Zhou et al. IndexTTS2, 2025. Li et al. EMORL-TTS, 2025. Ma et al. emotion2vec, 2023. Liao. TAT_MOE Corpus, 2022. Du et al. CosyVoice2, 2024.